

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

- A) Find out first SIX elements of the series 'R 10/3 (1,....., 1000)'. CO-1 6
- B) Explain, the basic design process in details. CO-1 6
- C) Write short note on 'Aesthetic Considerations in Machine Design' with suitable example. CO-1 6

Q.2 Solve Any Two of the following.

12

- A) Differentiate between "Maximum Shear Stress theory" and 'Maximum Distortion Energy Theory' CO-4 6
- B) The frame of a hydraulic press consisting of two steel plates is shown in figure 1. The frame is subjected to a maximum load of 20 kN. The plates are made of plain carbon steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$). The factor of safety is 2.5. Find the thickness of plates.

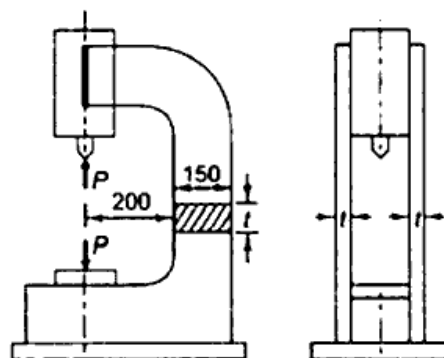


Figure 1

CO-2 6

- C) Design a Cotter joint subjected to a axial tensile load of 50 kN. All the components are made of plain carbon steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$). Use factor of safety 4 for all the components. CO-2 6

Q. 3 Solve Any Two of the following.

12

- A) Explain following terms:
 - a) Stress concentration
 - b) Notch Sensitivity
 - c) Endurance Limit
 CO-5 6
- B) A flat plate subjected to a tensile force of 5 kN is shown in figure 2. The plate material is grey cast iron FG200 ($S_{ut} = 200 \text{ N/mm}^2$) and the factor of safety is 2.5. Determine the thickness of the plate. CO-5 6

Assume: for hole section $K_t = 2.16$, for fillet section $K_t = 1.8$

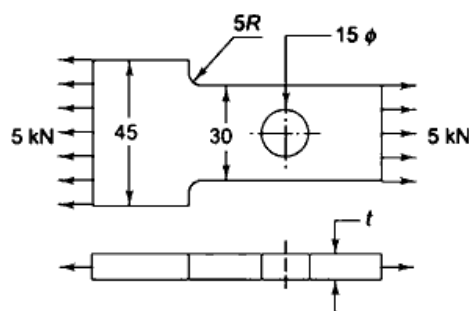


figure 2

- C) Differentiate between Soderberg and Goodman lines.

CO-5

6

Q.4 Solve Any Two of the following.

12

- A) What is feather key? Give its applications.

CO-6

06

- B) A rotating shaft 40 mm in diameter, is made of plain carbon steel FeE ($S_{yt} = 580 \text{ N/mm}^2$). It is subjected to a steady torsional moment of 250 N-m and bending moment 1250 N-m. Calculate the factor of safety based on,

CO-6

06

- Maximum principal stress theory
- Maximum shear stress theory

- C) Write short note on 'ASME code for designing transmission shaft'.

CO-6

06

Q. 5 Solve Any Two of the following.

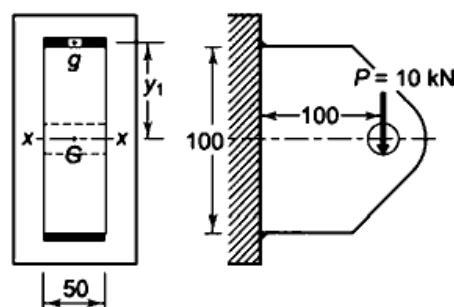
12

- A) Derive expression for torque required for lowering the load using power screw.

CO-6

6

- B) A bracket is welded to the vertical column by means of two fillet welds as shown in figure 3. Determine the size of the welds, if the permissible shear stress in the weld is limited to 70 N/mm^2 .



CO-6

figure. 3

6

- C) • Explain the step-by-step procedure for designing the helical springs.

CO-6

6

*** End ***